

# 05

## The Plan, Explained

The approved plan, section by section, with a comment on each

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Understand the project.

Know what deserves your attention.

### COLORFUL READING EDITION

The complete original text, with reading maps, priority labels and selective highlights. Start with the map on the next page.

## Your reading map

### Your decisions, responsibilities, costs, and next steps

Start with the links below. Then read the full text at your own pace.

- **Scope of the approval** - Distinguish the reading stage from the build.
- **The five next steps** - Follow the order and the owners.
- **The decisions** - See what is chosen and what needs your input.
- **Who does what** - Connect your role to the build and review work.
- **Milestones** - Understand each stage's finish line.
- **Budget** - Read the ceiling and timing of the costs.
- **Risks and cut lines** - See what protects the project when plans change.

**Keep for later.** The vendor scorecard and service details are for setup. The technical appendix is a later reference. Shaded comments explain what each section means for you.

### The four labels

| Label                        | How to use it                                                          |
|------------------------------|------------------------------------------------------------------------|
| <b>READ FIRST</b>            | Blue. Essential to understanding the project or a decision.            |
| <b>KEY IDEA</b>              | Teal. A concept or checkpoint to remember; yellow marks short phrases. |
| <b>WATCH OUT</b>             | Orange. A limitation, assumption, or common mistake to read carefully. |
| <b>OPTIONAL - FIRST PASS</b> | Purple. Return when you need this detail; the text is still here.      |

*Reading labels are added guidance. The original text, figures, dates, and source references are retained. Page numbers in this edition may differ from the originals.*

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## How to read this document

This is document 5 of the five I promised you. The first four were delivered under M0 (plan section 3): 1-read-this-first.pdf, 2-damodaran-essentials.pdf, 3-the-other-side.pdf and 4-the-guide.pdf. A PDF is a fixed-layout document file. This fifth one walks through the approved plan, section by section, with my comment on each part.

**What the plan is.** The plan is a written file, version 4, dated 2026-09-14. A milestone is a named stage of the work with a clear finish line. The plan has six milestones, M0 to M5 (plan section 5). Only M0 is approved. M0 is the documents and diagrams, and I finished it today. Everything after M0 is parked, meaning agreed in direction but frozen until you say “go”. Nothing after M0 has been built: no repository work, no accounts, no installs. A repository, or repo, is the folder where code lives with its history.

**How each chapter is shaped.** Every chapter after this one has the same three parts.

- **In short.** A few plain sentences that restate that part of the plan, with every term defined on first use.
- **Comment.** An indented note with four fixed lines. *What it means for you* says what changes for you. *Decision needed* names one of your three decisions, or says none. *Money* gives the cost from the budget (plan section 6), or none. *When* gives the milestone and its calendar words (plan section 5).
- **Read more.** The PDF and chapter that go deeper.

**Reading order.** Read 1-read-this-first.pdf first. Read 2-damodaran-essentials.pdf before this document or alongside it. On your first pass, skip the chapter on Section 10. Section 10 holds the technical specifications, meaning exact descriptions of how each piece is built. One example is the tables of the database, the one store for the numbers (plan section 10.3, named in plan section 5). It will make more sense once the foundation exists.

**Your three decisions.** These are the only decisions the plan asks of you. Everything else is already decided.

1. Go or wait. Say “go” only after you have read PDF 1 and PDF 2. Nothing after M0 starts before that word (plan section 0).
2. Which 10 to 20 US companies go on the first watchlist, the list of companies the system values and re-values. I will need the first names during M2, days 3 to 7 after go, and 20 names by the end of M3, weeks 2 to 4 (plan section 5).
3. Yes or no to opening the free accounts in plan section 7: GitHub, OpenRouter, Neon, Pinecone, Voyage, Cloudflare, Langfuse and Hex. Mistral and EODHD, two paid services, come later (plan section 6). This happens in M1, three evenings, only after go (plan section 5).

**Where the full plan lives.** The complete text is on your Mac at plans/help-me-out-here-abstract-wilkinson.md. This document restates it in shorter sentences. When the two differ, the plan file is the source.

## Section S: Scope of this approval, M0 only

### READ FIRST

**Start with what this document says is approved.** Separate the reading stage from the later build stages, and keep that boundary in mind through the rest of the plan.

**In short.** The plan runs in stages called milestones, and M0 is milestone zero, the reading stage (plan section 5). You asked to read and think before anything gets built, so this approval covers M0 only (plan section S). M0 means four documents, the FigJam diagrams, and the accounts checklist (plan section S). Figma is a drawing tool in the browser, and FigJam is its whiteboard for diagrams. The accounts checklist lists the free online services you would sign up for later. Everything after M0 stays parked in the plan until you say “go” (plan section S). In this pass nothing was installed, no repo was created, no account was opened, and no code was written (plan section S). A repo is a repository, a folder of code that remembers every change. The five M0 deliverables, in the order I produced them, are all delivered today, 2026-09-14 (plan section S):

- Document C, the Damodaran revision brain-dump, written from your lecture notes with a pointer to the source behind each idea (plan section S). Delivered as 2-damodaran-essentials.pdf.
- Document A, the system in plain words, with the diagrams, the accounts checklist and the correctness rules (plan section S). Delivered as 1-read-this-first.pdf.
- Five FigJam boards: architecture, nightly data flow, the valuation chain, the memo pipeline and the milestone map (plan section S). Delivered in your Figma account.
- Document B, the teaching guide, with a Figma chapter and a two-hour starter path (plan section S). Delivered as 4-the-guide.pdf.
- Document D, the critics reader (plan section S). Delivered as 3-the-other-side.pdf.

All of it sits in the docs/ folder inside ~/Valuation/ on this Mac (plan section S). One note on Figma. The Figma connector, a link that lets me act inside Figma for you, is signed in as you on the starter plan with a “View” seat, the most basic license (plan section S). The five boards exist, and their PNG images, plain picture files, are also inside 1-read-this-first.pdf. So opening Figma is optional for now.

### Comment

**What it means for you:** Nothing on your Mac has changed except a folder of documents. You can read at your own pace, and the rest of the plan waits.

**Decision needed:** none for this section, it is already approved. The next one is decision (1), go or wait, after you read 1-read-this-first.pdf and 2-damodaran-essentials.pdf.

**Money:** none. M0 cost nothing new; the Cursor Pro \$20 per month was already being paid before this plan (plan section 6).

**When:** M0, documents and diagrams, done by me today, 2026-09-14 (plan section 5).

**Read more.** 1-read-this-first.pdf, section “14. What happens next”, explains what waits for your “go”, and its section “10. The accounts you will create, in order” is the accounts checklist. 4-the-guide.pdf, “Chapter 13. Figma and FigJam: the whiteboard where the system is drawn”, teaches the boards.

## Section 0: What happens next, in five steps

### READ FIRST

**Read the five steps as a sequence.** Notice the owner and timing of each step; the next chapters explain them in more detail.

**In short.** This section is the whole plan as five steps, each with an owner and a rough length.

1. I write the four PDFs and the diagrams, which are pictures of the system drawn in FigJam, an online whiteboard. This step is already done, today, 2026-09-14 (plan section 5, M0).
2. You will read 1-read-this-first.pdf and 2-damodaran-essentials.pdf (the plan calls them documents A and C), and open the accounts on the checklist in PDF 1. This will take 3 evenings, and only after you say “go” (plan section 0). No code yet.
3. I will build the foundation as code in your private repo, the GitHub folder that holds all the code. That means the calculator, the database, the librarian that finds Damodaran passages, the memo writer, the tools Cursor calls, the overnight robots and a first dashboard. I will also write a walkthrough explaining every file. Days 3 to 7 after go (plan section 0).
4. You will work in Cursor, a code editor with an AI helper built in, one guide chapter and one small change at a time. Weeks 2 to 4, with me on call (plan section 0).
5. India, the accuracy scoreboard, ranges and the market panel will follow in weeks 4 to 8, you driving, me on request (plan section 0).

You asked about “the other md files”. An md file is a Markdown file, plain text with simple marks for headings. Documents A to D, the four PDFs including the guide, are reading, and so is the walkthrough. Everything else from step 3 is code and settings in the repo, which you will change through Cursor.

### Comment

**What it means for you:** Only step 1 is finished; nothing else starts until you say “go”, so read at your own pace.

**Decision needed:** Decision 1, go or wait, after reading 1-read-this-first.pdf and 2-damodaran-essentials.pdf; decision 3, the free accounts, comes in step 2 after go.

**Money:** Cursor Pro \$20 per month, already paid; in step 2 the first OpenRouter purchase, \$20 of credits with a \$75 spend limit (plan section 6).

**When:** M0 done today 2026-09-14; then M1, 3 evenings after go; M2, days 3 to 7 after go; M3, weeks 2 to 4; M4, weeks 4 to 8 (plan section 5).

**Read more.** 1-read-this-first.pdf, section “14. What happens next”, and its section “10. The accounts you will create, in order” for step 2. 4-the-guide.pdf, Chapter 16 (The schedule: twelve evenings-only weeks, from “go” to a working system), goes deeper on the calendar.

## Section 1: What we are building, in one page

### KEY IDEA

**Connect this to PDF 1.** Follow the seven parts and the comment underneath them, especially the distinction between exact arithmetic and uncertain inputs.

**In short.** The first section of the plan describes the whole system in seven parts (plan section 1). The goal is a machine that does what you used to do by hand in Damodaran's spreadsheet, for many companies at once, in India and the US, private to you, with nothing on your Mac except an editor (plan section 1).

- **The calculator.** A program that takes a company's numbers and Damodaran's reference tables and computes value per share the way his spreadsheet does (plan section 1). Think of a pocket calculator checked against an answer key. The golden test is that answer key: the spreadsheet ships with a worked example, Almarai, a Saudi food company, valued at 7.187840270062114 per share, and the program must reproduce that number to six decimals before it may value anything else (plan section 1). "Exact" covers only the arithmetic, including how an industry beta (how much a stock swings with the market) is adjusted for the company's debt; it says nothing about whether the inputs are right (plan section 1). So every valuation will also produce a range from Monte Carlo draws (many reruns with varied inputs) and a sensitivity table (a grid showing how the value moves when one input at a time is changed) (plan section 1).
- **The database in the cloud (Neon).** Neon is a hosted database, a filing cabinet on the internet (plan section 1). Every number will carry three labels: source, license class and vintage (plan section 1). License class means whether you may show the number to other people (plan section 1). Vintage means which snapshot of Damodaran's data the number belongs to; he updates his tables every January, with quarterly and monthly refreshes of two of them, so one year holds several versions of the risk-free rate and the risk premium that must never be mixed (plan section 1). The database will refuse a number that is missing any label (plan section 1).
- **The librarian in the cloud (Pinecone and Voyage).** His 609 blog posts, lecture packets, book, datasets and transcripts will be cut into about 60,000 passages (plan section 1). Each passage becomes a vector, a list of numbers that captures its meaning, so a question can find the right passages, and answers cite them (plan section 1). Picture a librarian who hands you the exact page, not the whole book.
- **The memo writer.** For each company an AI model will read the whole latest filing in one go, up to about 1.3 million tokens (a token is a small chunk of text), plus the matching Damodaran passages (plan section 1). It will write a "story to numbers" memo in his own template, proposing each calculator input with a citation to a filing page or a Damodaran passage (plan section 1). Think of a junior analyst who must footnote every claim. The 3P test sorts each claim as possible, plausible or probable; only probable claims may set a base-case input, plausible ones set a scenario, and possible ones are recorded at zero (plan section 1). Reverse DCF asks what growth or margin the current share price already requires (plan section 1). The memo must also name the company's uncertainty type from his "Dark Side of Valuation", such as young, distressed, cyclical, financial-service, complex holding or emerging-market, and write down the failure probability and the expected range (plan section 1). You review, edit and approve; only then does the calculator run (plan section 1). A second, opposing memo per company is

mandatory, the “Gurley field”, named after the Uber episode where he re-ran his model on Bill Gurley’s story (plan section 1).

- **Robots that run at night (GitHub Actions).** GitHub Actions is a scheduler that runs small jobs on rented computers, like a night watchman doing rounds (plan section 1). The jobs will pull new filings from SEC EDGAR, the free public archive of US filings, and later from EODHD, a paid service, for India (plan section 1). They will re-value your watchlist, check for a new Damodaran risk-premium file, pull licensed headlines, and write it all to the database (plan section 1). “Nightly, never intraday” means once a night is the right pace, and nothing here reacts to prices during the trading day (plan section 1).
- **The private dashboard (Hex).** Hex is a hosted notebook, a private wall of charts over the same database (plan section 1). It will show your watchlist by sector and country, value versus price, Monte Carlo ranges, the accuracy scoreboard, and a descriptive “is the market on the rails” panel built from Damodaran’s own implied ERP series (ERP is the equity risk premium, the extra return investors demand for holding stocks rather than government bonds) (plan section 1). The scoreboard measures the bias (whether past valuations ran high or low on average) and the variance (how widely they scattered) of past valuations against realized prices (plan section 1). You look at it; you do not print it (plan section 1).
- **Cursor as your daily tool.** Cursor is the editor you already pay for; the system’s tools plug into it as an MCP server, a standard socket that lets an editor call outside tools (plan section 1). You will type requests such as “value Trent” or “write the memo for Apple” (plan section 1). Cline, which is free, sits in the same window as a bench for trying open-weight models (models whose files are published, so anyone can run them) (plan section 1). Think of Cursor as the dashboard of a car whose engine lives elsewhere.

#### Comment

**What it means for you:** Nothing runs on your Mac; the calculator, database, librarian, robots and dashboard live in the cloud, and you only type in Cursor. The Almarai number proves the arithmetic; everything after it is an estimate with a range.

**Decision needed:** Decision 1, go or wait, after you read 1-read-this-first.pdf and 2-damodaran-essentials.pdf. Later, decision 2, which 10 to 20 US companies go on the first watchlist, because the robots and the dashboard need names to work on.

**Money:** About \$120 to 200 per month for the whole system, ceiling \$200 (plan section 6). Today only Cursor Pro at \$20 per month is being paid, and it is already paid (plan section 6).

**When:** M0, documents and diagrams, me, done today 2026-09-14, describes it. M2, foundation build, me, days 3 to 7 after go, will build the first version of all seven parts. M4, weeks 4 to 8, will add India, the scoreboard and the market panel (plan section 5).

**Read more.** In 1-read-this-first.pdf, “1. What we are building, in one page” and “3. The six rules that make the numbers trustworthy”, including “3.1 The golden test”, “3.2 Vintage” and “3.3 Source and licence class”. In 2-damodaran-essentials.pdf, “Chapter 1: Why value, the story, and the 3P test” and “Chapter 5: Growth, Terminal Value, and the Dark Side”. In 3-the-other-side.pdf, “The Other Side, Chapter 4: Twelve Episodes” for the Uber and Gurley story behind the opposing memo. In 4-the-guide.pdf, “Chapter 2. Cursor and Cline: the tool you type into, and the free bench beside it”, “Chapter 3. GitHub Actions: the robot that tests your code and runs it at night”, “Chapter 6. Pinecone, Voyage and RAG in plain words” and “Chapter 12. Hex: a private dashboard over Neon”.

## Section 2: Your decisions, and the two that changed after your review

### READ FIRST

**Focus on what is already chosen and what needs your input.** Read the table and its comments together before treating every listed item as a new decision.

### The decisions table

**In short.** This part of the plan records **eight decisions**, and two of them changed after your review of the earlier draft (plan section 2). Here is each row in plain words.

- **Cursor, not pi, as the daily tool.** Cursor is a code editor with an AI helper built in, and you already pay for it. It shows every change as a diff, a before-and-after view of a file, and its Plan mode shows an English plan before any change. pi is a bare terminal tool, a text-only command window, with no safety gates, so it is not a first tool. Cline, a free add-on inside Cursor, will be the bench for trying other models. You will not buy a second paid tool (plan section 2).
- **Open-weight models in the runtime, not the editor.** An open-weight model is one whose trained parameters are published, so anyone can run it. The runtime is the part of the system that runs by itself at night; the editor is the tool you type code into. Putting your own model key into Cursor is unofficial and breaks its privacy promise, so Cursor will keep its own models. Open-weight models will run inside your system, through OpenRouter, one account that reaches many models (plan section 2).
- **Outsource everything.** Every part will be a hosted service, like hiring specialists instead of building a workshop: Neon for the database, Pinecone for search, Voyage for turning text into searchable numbers, OpenRouter for models, GitHub for code and night jobs, Langfuse for model logs, Cloudflare R2 for raw files, Hex for dashboards, Mistral for scanned pages, EODHD for India data. You will manage no servers (plan section 2).
- **Private, always.** The code will live in a private repository, the folder GitHub keeps code in, inside a GitHub organization you own; an organization is a shared workspace that keeps faster job machines available later. No public web app, and Cursor Privacy Mode stays on (plan section 2).
- **Money is not the object, time is.** The ceiling is \$200 per month. I will build the foundation in the first week while you create accounts and read the four documents; from week two you will drive Cursor (plan section 2).
- **Explain everything.** Four documents and diagrams come first, and every milestone says what you will see (plan section 2).
- **Worth it if someone else ever uses it.** Every stored number will carry a license label, a tag saying where it came from and what you may do with it, so going public later would be a data license purchase plus web screens, not a rewrite (plan section 2).
- **Earlier decisions kept.** US first, then India; a watchlist of 20 to 50 names, where a watchlist is the list of companies the system follows; research only; no tracking of holdings; a descriptive market panel allowed, advice not (plan section 2).

**Comment**

**What it means for you:** The tool you type into stays the one you already know, and the clever model routing happens out of sight, inside the system.

**Decision needed:** Decision 1, go or wait, after you read 1-read-this-first.pdf and 2-damodaran-essentials.pdf; the eight rows themselves are already decided in the plan.

**Money:** Cursor Pro \$20 per month, already paid; the overall ceiling is \$200 per month (plan section 6).

**When:** M0, documents and diagrams, me, done today 2026-09-14; the Cursor and Cline setup comes in M1, you, 3 evenings, starting only when you say go (plan section 5).

**Read more.** In 4-the-guide.pdf, read “Chapter 2. Cursor and Cline: the tool you type into, and the free bench beside it”. In 1-read-this-first.pdf, read “2. Where the pieces live and who runs them”.

## 2a. What the point of open-weight models is

**In short.** Open-weight models such as GLM, Kimi, Qwen and DeepSeek cost 7 to 40 times less per token than frontier models (plan section 2a). A token is a piece of a word, and models charge by the token; a frontier model is one of the most capable closed models of the moment. An open-weight model can be pinned to an exact checkpoint, one frozen version, so a valuation run in 2026 could be re-run in 2029 with the same model, which closed models cannot promise (plan section 2a). The licenses, the legal terms of use, are permissive: GLM-5.3-Flash uses the plain MIT license, and GLM-5.3 and Kimi K3 have revenue clauses that do not touch a private project (plan section 2a). The price is capability. On the Artificial Analysis Intelligence Index v4.3, a public scoreboard dated 2026-09-07, the frontier scores 53 and the best open-weight model, GLM-5.3, scores 45 (plan section 2a). That gap appears in long tasks where a model takes many steps alone, not in the summarizing and extracting that fills a valuation pipeline, the chain of steps from raw data to memo. One correction: I earlier wrote that Meta’s Muse Spark 1.3 was open-weight, and it is not (plan section 2a). The rule will be to pick a model per job by cost per task and hallucination rate, where a hallucination is a confident false statement, with every choice in one file that changes in one line (plan section 2a).

**Comment**

**What it means for you:** Think of hiring a reliable clerk for routine paperwork rather than a senior partner; the clerk is cheaper, and paperwork is most of the job.

**Decision needed:** none, already decided in the plan.

**Money:** OpenRouter \$30 to 70 per month, with a first purchase of \$20 of credits and a \$75 spend limit (plan section 6).

**When:** M1, you, 3 evenings after go, sets up the Cline bench; M2, foundation build, me, days 3 to 7 after go, wires the models into the runtime (plan section 5).

**Read more.** In 4-the-guide.pdf, read “Chapter 7. OpenRouter and benchmark literacy: one door to every model, and how to read the scoreboard”. In 1-read-this-first.pdf, read “4.1 The model table, explained line by line”.

## 2b. Plain answers to the questions from your review

**In short.** You asked five questions in your review, and the plan answers each one (plan section 2b).

- **Two places a model is used.** Place 1 is while you write code, in Cursor, on the models it includes. Place 2 is inside the product at night, when our own code asks a model to write memos, answer questions or tidy call transcripts. Open-weight models go in place 2, because

that is where the volume and cost are, and a pinned model keeps old valuations reproducible (plan section 2b).

- **The point of pi.** pi is for people who want to build their own coding tool from a few basic parts and pay the least per task. I would use it one day to run a scripted agent, a model that acts on its own, inside a nightly job. You need a valuation system, not a coding tool, so pi is deferred, not judged (plan section 2b).
- **Tavily and Firecrawl, GitHub Actions, Blacksmith.** These are three different things. Tavily and Firecrawl are search and crawl services paid per query, and on the Artificial Analysis Search Index of 2026-08-18 Parallel and Exa beat them on quality per dollar. GitHub Actions is the scheduler, the alarm clock that wakes at night and runs the jobs. Blacksmith is a faster machine for those jobs, irrelevant until you pass the 2,000 free minutes. Cursor Automations can also schedule agents, but they bill through Cursor and tie the pipeline to your editor, so they will only do small chores (plan section 2b).
- **Estimates yes, advice no.** The system will print value per share, price as a percentage of value, the chance that value is below price from Monte Carlo draws (many re-runs with random inputs), what the price already requires, the market's implied risk premium against its history, and a scoreboard of how past valuations fared after 90, 180 and 365 days (plan section 2b). The implied risk premium is the extra return the market demands for stocks over safe bonds, read from today's prices. It will never print advice words. "Is it working for the next month" cannot be answered: Damodaran's own data says no valuation method forecasts next year's return well, and one month is noise. The honest question is "is the method working so far", and the scoreboard will answer it as valuations age (plan section 2b).
- **Visit the librarian first.** Yes. The librarian is the part that answers your questions with citations from Damodaran's writing. 2-damodaran-essentials.pdf, which the plan calls document C, comes right after 1-read-this-first.pdf for that reason, and the librarian's first job in M2 will be answering your questions while you read (plan section 2b).

#### Comment

**What it means for you:** None of these answers adds to your workload; they settle where each tool sits, and the scoreboard, not a monthly feeling, will say whether the method is working.

**Decision needed:** none, already decided in the plan.

**Money:** GitHub CI, the robot that tests code on every change, \$0 to 25 per month; optional Parallel search about \$18 (plan section 6).

**When:** M2, foundation build, me, days 3 to 7 after go, for the librarian and the nightly scheduler; M4, you in Cursor, weeks 4 to 8, for the scoreboard (plan section 5).

**Read more.** In 1-read-this-first.pdf, read "4. The two places an AI model is used" and "6. Estimates, not advice: what the scoreboard shows". In 4-the-guide.pdf, read "Chapter 3. GitHub Actions: the robot that tests your code and runs it at night". In 3-the-other-side.pdf, "The Other Side, Chapter 2: His Misses with the Numbers" shows why one month tells you nothing.

## 2c. What v1, v2 and v3 mean

**In short.** v1, v2 and v3 are labels for scope, not a promise of exactly three versions (plan section 2c). v1 covers weeks 1 to 8 and means private, working and correct: the calculator with its golden test, the check that the engine reproduces the Almarai workbook answer; the database with vintages and licenses, where a vintage is the dated version of an input; the librarian; the memo writer for US names; the nightly robots; the first Hex dashboard (plan section 2c). v2 covers months 3 to 6 and

adds India, the accuracy scoreboard, Monte Carlo and reverse DCF everywhere, the market-regime panel, opposing memos, and whatever your coursework suggests, such as backtests over the 1998 to 2024 archives (plan section 2c). DCF is discounted cash flow, the method that values a company from its future cash; reverse DCF works backward from the price to the growth it assumes; a backtest runs the method on old data; the market-regime panel shows what the market as a whole is pricing in. v3 happens only if others use it, adding a web app, commercial data licenses, multiple users and a public method page (plan section 2c). Nothing built in v1 will be rewritten for v2 or v3 (plan section 2c).

#### Comment

**What it means for you:** You get a private, correct system first, and every later stage adds rooms to a house whose foundation is already finished.

**Decision needed:** Decision 2, which 10 to 20 US companies go on the first watchlist, because v1 memos are for US names; I will ask for the list after go, since M3 ends with a 20-name watchlist (plan section 5).

**Money:** EODHD \$60 per month, only from M4, when v2 starts India (plan section 6); v3 costs are not priced in the plan.

**When:** v1 runs from M1 (you, 3 evenings, after go) through M4 (you in Cursor, weeks 4 to 8); v2 follows in months 3 to 6 (plan section 2c); v3 is M5, only if others ever use it, later (plan section 5).

**Read more.** In 4-the-guide.pdf, read “Chapter 16. The schedule: twelve evenings-only weeks, from”go” to a working system”. In 1-read-this-first.pdf, read “14. What happens next”.

## Section 3: The four documents and the diagrams

#### OPTIONAL - FIRST PASS

**A reference to the original deliverables.** The file descriptions and original page counts below are retained; this reading edition has its own pagination.

**In short.** Plan section 3 lists four documents and one set of diagrams. I deliver all of them in milestone M0, the first checkpoint of the plan (plan section 5). Each document exists as a PDF, a file that looks the same on every screen. The plan labels the documents A to D. The numbers in the file names are the order in which you should read them, so letters and numbers differ on purpose. The page counts below come from the delivered files as they sit on disk today, 2026-09-14.

- **A. Read This First** became 1-read-this-first.pdf. It explains the whole system in plain words with pictures: every part, every outside service, the accounts you will open, the correctness rules, and how to set up Cursor, your coding editor (plan section 3). The delivered file has 18 pages.
- **B. Teaching Guide** became 4-the-guide.pdf. It tells you where and how to learn each tool, in order, with hours and links, and ends with a 12-week schedule (plan section 3). The delivered file has 104 pages.
- **C. Damodaran, the important parts** became 2-damodaran-essentials.pdf. Damodaran is the professor whose valuation method this system follows. The document is your revision notes: the story-to-numbers frame, the DCF chain, country risk with the India numbers, and his rules of thumb (plan section 3). DCF means discounted cash flow, valuing a business by its future cash, reduced for time and risk. The delivered file has 53 pages.

- **D. The Other Side** became 3-the-other-side.pdf. It collects his own warnings, his misses with the numbers, the academic critiques, twelve episodes where someone bet against him and won, and what this means for our product (plan section 3). The delivered file has 47 pages.
- **FigJam diagrams.** FigJam is an online whiteboard made by Figma, a browser design tool. There are five boards: architecture, nightly data flow, the valuation chain, the memo pipeline, and the milestone map (plan section 3). I create them through the Figma connector, a link between my tools and your Figma account, so you can open and edit them (plan section 3). If your view seat blocks creation, the same diagrams ship as images inside document A, where the links also live (plan section 3).

#### Comment

**What it means for you:** All four PDFs and the diagrams are delivered, so this section asks nothing new of you. Your only job now is to read PDF 1 and then PDF 2. Nothing after M0 starts until you say “go”.

**Decision needed:** none for this section. Reading PDF 1 and PDF 2 prepares you for decision 1, go or wait.

**Money:** none.

**When:** M0, documents and diagrams, me, done today 2026-09-14.

**Read more.** Start with 1-read-this-first.pdf, “How to use this document”, then its “14. What happens next”. For the diagrams, see 4-the-guide.pdf, “Chapter 13. Figma and FigJam: the whiteboard where the system is drawn”. 2-damodaran-essentials.pdf opens with “How to read this document”, and 3-the-other-side.pdf with “The Other Side: Read Me First”.

## Section 4: Who does what, and when

#### READ FIRST

**Follow who does what.** Separate your learning and review work from the foundation build, then connect the timing to the milestones.

**In short.** This section is the plan’s calendar. **The clock starts** on the day you say “go”, not today. Only M0, the documents and diagrams, is done (plan section 5).

- Week 1, me. I will build the foundation, M2, in a private repo you own, plus a walkthrough that explains every file (plan section 4). A repo is the folder holding the code and its change history. Private means only you can open it.
- Week 1, you, 3 evenings (plan section 4). You will open the accounts, with a \$75 spend limit on OpenRouter, the door through which the system rents AI models (plan section 4). You will set up Cursor, the code editor with an AI helper inside it: Privacy Mode, which stops Cursor storing your code; the MCP file, which lists the tools the helper may call; and Plan mode, where the helper plans and waits (plan section 4). You will read 2-damodaran-essentials.pdf and the first three chapters of 4-the-guide.pdf (plan section 4).
- Weeks 2 to 4, you in Cursor (plan section 4). You will learn by changing what exists: the 10-minute exercises, your first pull requests, adding companies to the watchlist, reviewing memos, and reading the dashboard (plan section 4). A pull request is a proposed change that is checked before it joins the main code. The watchlist is the list of companies the system values. A memo is the written valuation the system drafts for your approval. One evening a week you will work with the AI helper off (plan section 4). I will be on call when something blocks you.

- Weeks 4 to 8, you in Cursor, me on request (plan section 4). India through EODHD, a paid data service for Indian financials; the sector-exposure and accuracy views; evals in Langfuse, tests of the AI's answers and the tool that records them; Monte Carlo, which reruns a valuation many times to give a range; reverse DCF, which works back from today's price to the growth it assumes; opposing memos; and the market-regime panel, which describes the market's state and never tells you what to do (plan section 4).
- Later, only if others use it: a web app, commercial data licenses, and sub-agents, small AI helpers that run alone (plan section 4).

#### Comment

**What it means for you:** It is a relay race. I run the first leg, you run the rest with me beside the track, and nothing starts until you say go. Your week 1 is setup and reading, not coding.

**Decision needed:** Decision 1, go or wait, after reading 1-read-this-first.pdf and 2-damodaran-essentials.pdf; decision 3, the free accounts, in M1 after go; decision 2, the first 10 to 20 US companies, in weeks 2 to 4.

**Money:** Cursor Pro \$20 per month, already paid; OpenRouter first purchase \$20 of credits, \$75 spend limit; EODHD \$60 per month only from M4 (plan section 6).

**When:** M0 done today, 2026-09-14; M1 you, 3 evenings, starts only when you say "go"; M2 me, days 3 to 7 after go; M3 you, weeks 2 to 4; M4 weeks 4 to 8; M5 later, only if others ever use it (plan section 5).

**Read more.** 4-the-guide.pdf, "Chapter 16. The schedule: twelve evenings-only weeks, from "go" to a working system", gives the evening-by-evening version. 1-read-this-first.pdf, "10. The accounts you will create, in order", is your week 1 checklist.

## Section 5: Milestones

#### READ FIRST

**Read each finish line.** The checkpoint is more useful than the date alone: it tells you what must be true before the next stage.

### M0. Documents and diagrams

**In short.** A milestone is a checkpoint with a **clear finish line**. Plan section 5 has six, M0 to M5. M0 is my job, done today, 2026-09-14 (plan section 5). It delivers the four PDFs, the diagrams, and the accounts checklist. The diagrams live in FigJam, the whiteboard tool inside Figma. The checklist lists the services you will sign up for in M1. M0 is done when the four PDFs open, every link works, the FigJam boards open in your Figma account or their pictures sit in document 1, and the checklist is complete (plan section 5).

#### Comment

**What it means for you:** Nothing to build. Read PDF 1 and PDF 2, then tell me "go" or "wait".

**Decision needed:** Decision 1, go or wait, after PDF 1 and PDF 2.

**Money:** none.

**When:** M0, documents and diagrams, me, done today 2026-09-14.

**Read more.** 1-read-this-first.pdf, "1. What we are building, in one page" and "14. What happens next".

## M1. Accounts and Cursor

**In short.** M1 is yours: about 3 evenings, starting only when you say “go” (plan section 5). The steps, in plan order:

- GitHub stores code and its history. Turn on 2FA, two-factor authentication, a second code beside your password, and save the recovery codes offline. I will create the organization and the private repo, one project’s folder of code and history, and hand you ownership. Turn on secret scanning and push protection, which stop keys landing in the repo by mistake (plan section 5).
- OpenRouter is one door to many AI models with one key. Buy \$20 of credits, set a \$75 per month spend limit, turn prompt logging off, and turn zero-data-retention routing on, so providers keep no copy of what you send (plan section 5). Put the key in `~/ .zshrc`, the file your terminal reads at start; `echo $OPENROUTER_API_KEY | cut -c1-6` should print `sk-or-` (plan section 5).
- Free accounts, one project each named valuation: Neon, a hosted Postgres database; Pinecone Starter, a search index; Voyage, free for 200 M tokens; Cloudflare, one R2 storage bucket; Langfuse Hobby; Mistral pay-as-you-go; Hex Community (plan section 5). Each key goes into the repo’s `.env` file, never committed, and into GitHub Actions secrets, using the names in plan section 7.
- Cursor: turn Privacy Mode on in Settings, so Cursor stores none of your code. Open the repo. I will ship `.cursor/mcp.json`, which shows the system’s tools to Cursor’s agent, and `.cursor/rules/`, the hard rules. Try Plan mode (Shift+Tab): ask “explain how value\_company works”, read the plan, do not run it (plan section 5).
- Cline, optional and free, is a VS Code extension inside Cursor. Add the OpenRouter key and set GLM-5.3 for Plan and GLM-5.3-Flash for Act (plan section 5). It is your free bench for trying open-weight models before one goes into a robot.
- Day-one exercise: ask Cursor why terminal WACC equals the risk-free rate plus the mature ERP. WACC, the weighted average cost of capital, is the blended rate a company pays lenders and shareholders. ERP, the equity risk premium, is the extra return people ask for holding stocks instead of safe bonds. Then find the cell label Mature Market ERP + in the workbook yourself and compare (plan section 5).

M1 will be done when every key is in place, Cursor’s agent lists the valuation tools, and you have read document C, which is 2-damodaran-essentials.pdf (plan section 5).

### Comment

**What it means for you:** The only hands-on steps before you touch code. Two settings matter most: OpenRouter with spend limit \$75, logging off, zero-data-retention on; and Cursor Privacy Mode on.

**Decision needed:** Decision 3, yes or no to opening the free accounts listed in plan section 7, during M1 after go.

**Money:** OpenRouter first purchase \$20 of credits, spend limit \$75; Cursor Pro \$20 per month, already paid; other accounts \$0 (plan section 6).

**When:** M1, accounts and Cursor, you, 3 evenings, starts only when you say “go”.

**Read more.** 1-read-this-first.pdf, “10. The accounts you will create, in order”; 4-the-guide.pdf, “Chapter 2. Cursor and Cline: the tool you type into, and the free bench beside it”.

## M2. Foundation build

**In short.** M2 is my build, days 3 to 7 after go, while you read the walkthrough, a file that explains every folder, every command, and what to try first (plan section 5). The pieces:

- The repo: a private repository named `valuation` in TypeScript, holding `AGENTS.md`, the eight hard rules that Cursor, Cline and Claude all read (plan section 5). CI, continuous integration, is a robot that runs the tests every time code changes; it will use GitHub's Linux machines, 2,000 free minutes a month, then \$0.006 per minute, capped at \$25 (plan section 5).
- The calculator, `packages/engine`: a formula-by-formula port of Damodaran's spreadsheet. The golden test is one worked example with a known answer, Almarai at 7.187840270062114 per share against a price of 72.28, which the engine must match within 1e-6, that is to six decimal places (plan section 5; 1-read-this-first.pdf section 3.1). Its inputs will be pulled from the workbook by script, never typed, and CI will refuse code that fails it. Reverse DCF asks what growth or margin today's price assumes, and Monte Carlo draws many random inputs to give a range.
- The database, `packages/db`, on Neon: the tables from plan section 10.3, with constraints proven, for example a number without a vintage is refused. A vintage is the date stamp of the data file a number came from (plan section 5).
- US ingest: filings from SEC EDGAR, the US regulator's free filing site, for Apple, Costco and three more US names you choose (plan section 5). Apple's missing interest-expense line will show as a visible flag, never a silent zero.
- The librarian, `packages/rag`: RAG, retrieval-augmented generation, means the model looks up passages before answering. Blog posts and dataset sentences will sit in Pinecone, and both smoke questions, two check questions with known answers that prove the librarian works end to end, must pass (plan section 5).
- The memo writer, `packages/memo`: it will read the filing, retrieve Damodaran passages, and write the "Stories to Numbers" table plus a claims table; an uncited claim will reject the memo. A cost of capital outside 5.3% to 9.9% for the US, or 6.3% to 11.7% globally, will be flagged, and you approve in Cursor before a run (plan section 5).
- The MCP server, `packages/mcp-server`: MCP, Model Context Protocol, is a standard way to hand tools to an AI assistant. It will expose `value_company`, `write_memo`, `review_memo`, `get_industry_stats`, `search_corpus` and `report` from Cursor, Cline, Claude and pi alike (plan section 5).
- Nightly robots are scheduled jobs that run while you sleep: an EDGAR refresh, a watchlist revaluation, a watch that opens a GitHub issue when Damodaran posts a new file, and a licensed headline pull (plan section 5).
- The first Hex dashboard over Neon: watchlist, exposure, value-versus-price distribution, and the Almarai and Apple runs with ranges (plan section 5).

M2 will be done when `value_company AAPL US` prints value per share, WACC, rating basis, vintage and license; `write_memo AAPL` produces a cited memo; `search_corpus` answers the smoke questions with citations; the golden test is green in CI and a deliberate change turns it red; the nightly job has run once; and the Hex dashboard shows the watchlist (plan section 5).

#### Comment

**What it means for you:** You read and try commands; you do not write code yet. If I break the calculator, the golden test will turn CI red before it reaches you.

**Decision needed:** Decision 2 starts here: I will need three US names beyond Apple and Costco, the seed of your 10 to 20 company watchlist.

**Money:** OpenRouter \$30 to 70 per month; GitHub CI \$0 to 25; Neon, Pinecone, Voyage, Langfuse and

Cloudflare R2 \$0 (plan section 6).

**When:** M2, foundation build, me, days 3 to 7 after go.

**Read more.** 1-read-this-first.pdf, “3.1 The golden test” and “7. The nightly robots”; 4-the-guide.pdf, “Chapter 3. GitHub Actions: the robot that tests your code and runs it at night”.

### M3. You take the wheel

**In short.** In M3 you drive, in Cursor, weeks 2 to 4 (plan section 5). You will follow 4-the-guide.pdf in order. Each chapter ends with a 10-minute exercise on the real repo (plan section 5): resolve a merge conflict, two edits to the same line that Git cannot combine alone; break the golden test and watch CI go red; add a company to the watchlist; edit a memo; add a diagnostic warning; re-index one blog year; add a chart in Hex. You will track lines you wrote against lines you accepted from the tool (guide chapter 0). M3 will be done when you have merged five PRs, pull requests, which are proposed changes reviewed before merging, the watchlist holds 20 names with approved memos, and you can explain every rule in AGENTS.md out loud (plan section 5).

#### Comment

**What it means for you:** The learning phase. Every exercise is small, but each touches the real system, so the skills stick.

**Decision needed:** Decision 2, which 10 to 20 US companies go on the first watchlist; the finish line is 20 names with approved memos (plan section 5).

**Money:** Cursor Pro \$20 per month, already paid; OpenRouter \$30 to 70 per month (plan section 6).

**When:** M3, you take the wheel in Cursor, weeks 2 to 4.

**Read more.** 4-the-guide.pdf, “Chapter 0. How to learn with an AI coding tool without becoming dependent on it” and Chapter 16, The schedule: twelve evenings-only weeks, from “go” to a working system.

### M4. India, accuracy, ranges, the market panel

**In short.** M4 runs weeks 4 to 8, you in Cursor, me on request (plan section 5). The pieces:

- EODHD is a paid vendor of company financials through an API, a machine-readable feed. Check its free tier on TRENT.NSE, the retailer Trent on India’s National Stock Exchange, then subscribe to Fundamentals at \$59.99 per month, or use indianapi.in if it is as deep for less (plan section 5). Cross-check once against Trent’s own annual report.
- The INR risk-free rule: the rupee risk-free rate is the RBI 10-year G-sec yield, a 10-year Indian government bond, entered monthly as a vintage row, minus India’s default spread, the extra yield India pays over a top-rated borrower (plan section 5). In the January 2026 vintage the spread is 1.87% and total ERP is 7.08%, the mature ERP 4.23% plus a country risk premium of 2.85% (2-damodaran-essentials.pdf chapter 7). Industry averages will fall back from India to emerging markets to Global below 10 firms, recorded on every run (plan section 5).
- Sector-exposure and value-versus-price views for 20 to 50 India plus US names, research only; a test will forbid advice words (plan section 5).
- Langfuse will trace every model call; you will do error analysis on 100 real memos and answers (plan section 5).
- The accuracy scoreboard: realised\_price at 90, 180 and 365 days (plan section 5). Bias is the average signed error of value against later price: does the engine run high or low. Variance

is the spread of those errors: how scattered they are. Both by sector, region, vintage and memo version. The banner will read “The engine is tested (Almarai 1e-6). The forecast is not.” (plan section 5).

- The market-regime panel, descriptive only, spec in plan section 10.7. Implied ERP is the equity premium backed out of today’s index price. The panel will show his monthly implied ERP against the 1960 to 2025 and 10-year history, expected return on stocks, earnings yield, company profits as a percentage of share price, against bond yield as context, high-yield spreads, the extra yield risky company bonds pay over government bonds, from FRED, the St Louis Fed’s free data site, top-7 concentration, how much of the index sits in its seven largest companies, and a fair-value grid at ERP 2% to 6%, each as a plain-language state with his own market-timing caveats printed (plan section 5). Headline sentiment is an annotation, never an input.

M4 will be done when `value_company TRENT IN` shows tier, the licence class of the data source, plus vintage, the exposure view renders, the first scored horizon appears after 90 days, and the panel reproduces his September 2026 numbers, implied ERP 4.09% and expected return 8.84% (plan section 5).

#### Comment

**What it means for you:** Indian names arrive, and the system starts grading its own past estimates; the first grade comes only after 90 days (plan section 5).

**Decision needed:** none, already decided in the plan.

**Money:** EODHD \$60 per month from M4; optional INDstocks API free, Kite Connect 500 rupees, Parallel about \$18; total about \$120 to 200 per month, ceiling \$200 (plan section 6).

**When:** M4, India, accuracy, ranges, market panel, weeks 4 to 8.

**Read more.** 2-damodaran-essentials.pdf, “Chapter 4: Discount Rates and Country Risk” and “Chapter 7: India and the Rules of Thumb”; 1-read-this-first.pdf, “6. Estimates, not advice: what the scoreboard shows”.

## M5. Only if others use it

**In short.** M5 has no date; it starts only if other people ever use the system (plan section 5). It would add a browser front end, a website instead of Cursor; app-level login; commercial data licenses, paid rights to show vendor data to others; sub-agents, helper AI programs that run tasks on their own; and a public method page. The plan sets no finish line, because the trigger, other users, may never arrive (plan section 5).

#### Comment

**What it means for you:** Nothing now. The private version is the whole plan until someone else asks to use it.

**Decision needed:** none, already decided in the plan.

**Money:** none now; the line that would move later is Hex, \$0 now and \$36 if Professional (plan section 6).

**When:** M5, only if others ever use it, later.

**Read more.** 1-read-this-first.pdf, section 5, What “scales”, in plain words; 4-the-guide.pdf, “Chapter 10. OCR, web crawling and sub-agents: three helpers, and when to leave them in the drawer”.

## Section 6: Budget at \$200 per month

### READ FIRST

**Read the ceiling, recurring lines, and one-off costs.** The comment explains how the original budget applies to you and when each cost appears.

**In short.** A budget is a list of what each tool will cost per month, with a ceiling you will not cross. The plan sets that ceiling at **\$200 per month** (plan section 6). Think of it as a row of small jars, most of them empty today, with one jar you only fill later. The table restates the plan's table with the same figures (plan section 6). "Later" means a paid tier you will only switch on if you need it.

| Line                                            | Now (plan section 6)            | Note (plan section 6)                                                                                                        |
|-------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Cursor Pro                                      | \$20 per month, already paid    | The code editor with an AI helper inside. Stays on Cursor's own models; no BYOK, meaning no "bring your own key".            |
| OpenRouter, runtime                             | \$30 to 70 per month            | One door to many AI models. "Runtime" means the models that write memos and answers while the system runs. Spend limit \$75. |
| EODHD Fundamentals                              | \$60 per month, only from M4    | Company financial data for India. I will test indianapi.in first; it may save about \$50.                                    |
| Hex                                             | \$0 now, \$36 later             | The dashboard tool. Community is free; Professional only for a published app with a daily refresh.                           |
| GitHub CI                                       | \$0 to 25                       | The robot that tests code on GitHub. 2,000 free minutes, then \$0.006 per minute, capped at \$25.                            |
| Neon                                            | \$0 now, about \$10 to 20 later | The database, the one place the numbers live. Free until 0.5 GB.                                                             |
| Pinecone                                        | \$0 now, \$20 later             | The searchable index of Damodaran's writing. Builder tier only if you re-index often.                                        |
| Voyage, Langfuse Hobby, Cloudflare R2           | \$0                             | Text embeddings, a log of every model call, and file storage. Free at this scale.                                            |
| Mistral OCR                                     | about \$10, one-off             | OCR reads text out of images: the three image-only decks and any scanned packet.                                             |
| Optional: Kite Connect, INDstocks API, Parallel | 500 rupees, free, about \$18    | India prices and news search. INDstocks API is free with an INDmoney account.                                                |
| Total                                           | about \$120 to 200 per month    | Ceiling \$200.                                                                                                               |

Hex, GitHub CI, Neon, Pinecone, Voyage, Langfuse and Cloudflare R2 all start at \$0 (plan section 6). EODHD is the one line that only starts at M4, the India phase, at \$60 per month (plan section 6). The total is about \$120 to 200 per month, under a ceiling of \$200 (plan section 6). Bloomberg Digital, your news and market reading subscription, is your own cost and sits outside this budget (plan section 6).

### Comment

**What it means for you:** Most of the stack will be free while I build it; the only new money before M4 is OpenRouter, at \$30 to 70 per month with a \$75 spend limit (plan section 6).

**Decision needed:** None now. The yes or no to opening the free accounts belongs to Section 7, and it comes in M1 after go.

**Money:** About \$120 to 200 per month in total, with a ceiling of \$200; Cursor Pro at \$20 per month is already paid (plan section 6).

**When:** M1, accounts and Cursor, you, 3 evenings after go, for the first \$20 of OpenRouter credits; M4, weeks 4 to 8, for EODHD at \$60 per month (plan section 6); paid tiers of Hex, Neon and Pinecone later, only if needed.

**Read more.** 1-read-this-first.pdf, section “12. Budget at up to \$200 a month”, shows the same table in a shorter form, and section “13. Bloomberg, INDmoney and news, honestly” covers what your Bloomberg subscription does and does not give the system. 4-the-guide.pdf, “Chapter 9. Data sources: where every number comes from, and what you may do with it”, goes deeper on EODHD and the other data lines.

## Section 7: Vendor scorecard and the accounts to create

### OPTIONAL - FIRST PASS

**For choosing and opening accounts.** Refer to the scorecard and setup checklist when you reach that stage.

### The scorecard: are they the best at what they do?

**In short.** A vendor is a company whose service the system will rent, such as a database or an AI model. Plan section 7 lists 14 roles, one pick each, the runners-up, and why the pick wins. Free tiers come first; paid lines appear only where nothing free is good enough.

- Coding tool: Cursor Pro. Already paid; its diff review (every change shown before you accept it) and Plan mode let a beginner learn instead of copying blindly. Runner-up: Cline, the free bench.
- Database: Neon, a hosted Postgres, the widely used open-source database. An idle database costs nothing, and cheap copies called branches allow safe experiments. Runner-up: Supabase Pro.
- Vectors: Pinecone Starter. A vector is a list of numbers that stands for the meaning of a text, so a vector store searches by meaning; free up to 2 GB (plan section 7). Runner-up: pgvector, an add-on that stores vectors inside Neon itself.
- Embeddings and rerank: Voyage. An embedding model turns text into a vector, and a reranker re-sorts search hits so the best come first; 200 million free tokens (plan section 7). Runner-up: Cohere Rerank 3.5.
- Model router: OpenRouter. A router is one account and one key that reaches many AI models, and OpenRouter alone enforces zero-data-retention routing, so the provider keeps no copy of your prompts. Runner-up: Vercel AI Gateway.
- Models: GLM-5.3 for memos and answers, GLM-5.3-Flash batch for bulk work, Kimi K3 as judge, a second model that grades the first (plan section 7). GLM has the lowest hallucination rate, the rate of confident false statements, among strong open-weight models; DeepSeek V4 Pro is excluded for a 94% rate (plan section 7). Runner-up: DeepSeek V4.1 Flash.
- Dashboard and notebooks: Hex, Community free, later Professional at \$36 per month (plan section 7). The only tool within budget with SQL against Neon, real Python, charts and private sharing. Runner-up: Deepnote at \$39 (plan section 7).

- Observability and evals: Langfuse Hobby. Observability records every model call so you can see what happened; evals are tests that score model output. Open source, with evals free. Runner-up: Braintrust.
- Job runner: GitHub Actions, the robot that runs code on a schedule or after each change. The code already lives there, so the pipeline does not depend on the editor vendor. Runner-up: Inngest.
- Raw file storage: Cloudflare R2. 10 GB free with zero egress, the fee some vendors charge to read files back out (plan section 7). Runner-up: Backblaze B2.
- OCR: Mistral Document AI 4.1. OCR turns a scanned page image into text; batch price \$2 per 1,000 pages, and best on the only independent 2026 OCR benchmark at 74.9%, tables 94.3% (plan section 7). Runner-up: Reducto.
- Web search, optional: Parallel. Top on quality, bottom on cost in the AA Search Index of 2026-08-18 (plan section 7). Runner-up: Exa; Tavily is rejected.
- India fundamentals: EODHD at \$59.99 per month (plan section 7). Fundamentals are the financial statement numbers, and EODHD alone documents long-history Indian statements in one tidy schema. Runner-up: indianapi.in.
- India prices and your holdings: INDstocks API from INDmoney, free, or Kite Connect at 500 rupees (plan section 7). Official and read-only; scraping the app is forbidden. Runner-up: NSE bhavcopy files.

#### Comment

**What it means for you:** Fourteen roles, fourteen picks, all chosen already (plan section 7); you will not compare vendors yourself.

**Decision needed:** none, already decided in the plan.

**Money:** Cursor Pro \$20 per month, already paid; every other line is \$0 today, and the full stack later runs about \$120 to 200 per month under a \$200 ceiling (plan section 6).

**When:** M0, me, done today 2026-09-14; the paid lines switch on later, EODHD only from M4, weeks 4 to 8.

**Read more.** 4-the-guide.pdf, “Chapter 7. OpenRouter and benchmark literacy: one door to every model, and how to read the scoreboard”, explains the model picks, and its “Chapter 9. Data sources: where every number comes from, and what you may do with it” covers the India feeds.

## Accounts to create, in order

**In short.** An account is a login at one vendor, and most come with an API key, a long secret string that lets code act as you. Each key will live in a `.env` file, a private text file on your computer, and in GitHub Actions secrets, the locked drawer the nightly robot reads from. Two-factor authentication (2FA) is a second check at login, usually a code from your phone. Plan section 7 lists 12 rows in order; the table restates them, and 1-read-this-first.pdf, section 10, has the click-by-click version.

| #  | Vendor                                                      | Plan               | Dollars per month<br>(plan section 7) | What it is for                   | Security note                                                                           |
|----|-------------------------------------------------------------|--------------------|---------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------|
| 1  | GitHub, plus a free organization                            | Free               | 0                                     | Code, history, nightly robot     | 2FA on; recovery codes kept offline; root of trust                                      |
| 2  | Cursor, your existing account                               | Pro                | 20                                    | Editor you type into             | Privacy Mode on; never paste keys into its BYOK settings                                |
| 3  | OpenRouter                                                  | Pay-as-you-go      | 30 to 70                              | One key, many models             | Spend limit \$75; logging off; zero-data-retention routing on; 2FA                      |
| 4  | Neon                                                        | Free, later Launch | 0 to 20                               | Database holding every number    | Sign up with email and password, not only GitHub; 2FA                                   |
| 5  | Pinecone                                                    | Starter            | 0                                     | Search by meaning                | 2FA                                                                                     |
| 6  | Voyage                                                      | Free               | 0                                     | Turns text into vectors          | Billing alert before the free grant ends                                                |
| 7  | Cloudflare                                                  | Free, R2           | 0                                     | Raw filings and decks            | Hardware key or TOTP; token scoped to one bucket, the named folder that holds the files |
| 8  | Langfuse                                                    | Hobby              | 0                                     | Records every model call         | 2FA                                                                                     |
| 9  | Mistral                                                     | Pay-as-you-go      | about 2, one-off                      | Scanned pages into text          | Use the batch endpoint, the cheaper queue for jobs that need not finish immediately     |
| 10 | Hex                                                         | Community          | 0, later 36                           | Private dashboard over Neon      | Private workspace; Neon connection set inside Hex                                       |
| 11 | EODHD, from M4                                              | Fundamentals       | 60                                    | Indian financial statements      | Read the redistribution clause; test 20 NSE tickers free first                          |
| 12 | Optional: INDstocks API, Kite Connect, Parallel, Braintrust | Various            | 0, about 6 (500 rupees), about 18, 0  | India prices, news search, evals | Broker tokens expire daily, so plan a re-login step                                     |

BYOK means “bring your own key”, and TOTP is a time-based one-time password, the six-digit code from an authenticator app. Plan section 7 puts Mistral at about \$2 one-off for the account, while plan section 6 budgets about \$10 one-off for all the scanning work.

#### Comment

**What it means for you:** In M1 you will open the free accounts (GitHub, Neon, Pinecone, Voyage, Cloudflare, Langfuse, Hex), plus OpenRouter and Mistral, which are pay-as-you-go, name one project valuation at each, and paste each key into `.env` and GitHub Actions secrets (plan section 5).

**Decision needed:** decision 3, yes or no to opening those accounts, taken in M1 after you say “go”; decision 1, go or wait, comes first, after you read PDF 1 and PDF 2.

**Money:** the first OpenRouter purchase is \$20 of credits with a \$75 spend limit (plan section 6); Cursor Pro \$20 per month is already paid; everything else in M1 is \$0.

**When:** M1, you, 3 evenings, starts only when you say “go”.

**Read more.** 1-read-this-first.pdf, “10. The accounts you will create, in order”, walks through every screen. 4-the-guide.pdf, “Chapter 2. Cursor and Cline: the tool you type into, and the free bench beside it”, covers row 2 and the free bench.

## Hygiene

**In short.** Hygiene here means the small habits that keep secrets out of the shared code and limit the damage if one leaks, like keeping the spare key with a neighbor rather than under the mat. Plan section 7 sets five rules.

- Never commit `.env`. To commit is to save a file into the repo’s permanent history, so a committed secret is a leaked secret. Commit `.env.example` instead, the same file with every value blank.
- Use a distinct email alias per vendor, an extra address that lands in your normal inbox, so a leak is traceable and one stolen login does not open the rest.
- Set hard spend caps at OpenRouter, Mistral and Parallel, so a runaway job stops at the cap.
- Rotate the GitHub token and the OpenRouter key every 90 days (plan section 7); rotating means issuing a new key and deleting the old one.
- Create a Neon branch before any migration the AI agent proposes. A migration changes the shape of the database tables, and a branch is a cheap copy, so a bad one can be thrown away.

#### Comment

**What it means for you:** Five habits, none technical; four are one-time settings in M1, and the fifth is a reflex whenever an agent proposes a database change.

**Decision needed:** none, already decided in the plan.

**Money:** none; the caps exist to protect the \$200 per month ceiling (plan section 6).

**When:** M1, you, 3 evenings after “go”, for the first four; the Neon branch rule starts in M2, me, days 3 to 7 after go, and stays for good.

**Read more.** 4-the-guide.pdf, “Chapter 14. Security hygiene for a solo builder”, covers every rule, and 1-read-this-first.pdf, “12. Budget at up to \$200 a month”, shows where the spend caps sit.

## Section 8: Bloomberg, INDmoney, OCR, crawling, sub-agents, benchmarks, Figma, fine-tuning

### OPTIONAL - FIRST PASS

**Detailed answers by service and tool.** Return to the relevant subsection when you need that capability; this chapter need not block the first pass.

### Bloomberg

**In short.** An API (application programming interface) is a door one program uses to ask another for data. Bloomberg has no consumer API in 2026; the download page you found connects to nothing without a paid Terminal contract (plan section 8). Its free headline RSS feeds, plain headline lists for reader programs, are a gray area under Bloomberg's terms (plan section 8).

#### Comment

**What it means for you:** The robot will store only title, link and time, keep them private, and never let a headline move a number (plan section 8).

**Decision needed:** none, already decided in the plan.

**Money:** none; Bloomberg Digital is your personal subscription outside the budget (plan section 6).

**When:** M2, foundation build, me, days 3 to 7 after go, in the nightly headline pull (plan section 5).

**Read more.** 1-read-this-first.pdf, section "13. Bloomberg, INDmoney and news, honestly".

### INDmoney

**In short.** Scraping means a program copying data off a page or app screen, and INDmoney's terms forbid it (plan section 8). Its broking arm publishes a free official API, the INDstocks API Suite, and the plan uses only its read-only endpoints for your own holdings and NSE prices, quotes from India's National Stock Exchange (plan section 8). It needs a KYC account, an identity-verified brokerage account, but no static IP, a fixed internet address (plan section 8).

#### Comment

**What it means for you:** Fundamentals, the accounting numbers, still come from EDGAR, the SEC's free filings database, Damodaran's tables and EODHD, a paid market-data provider (plan section 8).

**Decision needed:** none now; plan section 6 marks it optional.

**Money:** INDstocks API free with an INDmoney account; Kite Connect at 500 rupees is the alternative (plan section 6).

**When:** M4, India, accuracy, ranges, market panel, you in Cursor, me on request, weeks 4 to 8 (plan section 5).

**Read more.** 1-read-this-first.pdf, section "13. Bloomberg, INDmoney and news, honestly".

### Crawling Bloomberg or INDmoney at night?

**In short.** A crawler is a robot that visits pages on a schedule and brings text home; Firecrawl is one such service. It will not touch Bloomberg or INDmoney, because their terms forbid it (plan section

8). It may read Bloomberg's headline feeds, newsletters in your Gmail, SEC EDGAR's own feeds and search, and licensed search APIs such as Parallel and Exa (plan section 8).

#### Comment

**What it means for you:** Results become evidence lines for the memo writer and notes on the market regime panel, the market conditions dashboard, never inputs (plan section 8).

**Decision needed:** none, already decided in the plan.

**Money:** Parallel search is optional at about \$18; the rest is \$0 (plan section 6).

**When:** M2, foundation build, me, days 3 to 7 after go; regime panel notes in M4, weeks 4 to 8 (plan section 5).

**Read more.** 1-read-this-first.pdf, section "7. The nightly robots".

## OCR

**In short.** OCR (optical character recognition) turns a picture of a page into searchable text. Three of Damodaran's "Dark Side" decks are image-only, and only those three will go through OCR (plan section 8). pdffonts, a command that reports whether a PDF holds real text, checks first; Mistral Document AI's batch endpoint, the slower cheaper queue, does the work; every table must add up to its stated total (plan section 8).

#### Comment

**What it means for you:** Three decks, one job, and you will not run it yourself.

**Decision needed:** none, already decided in the plan.

**Money:** Mistral OCR about \$10, one-off (plan section 6).

**When:** not before M2, foundation build, me, days 3 to 7 after go; the Mistral account comes after the free M1 accounts (plan section 7).

**Read more.** 4-the-guide.pdf, "Chapter 10. OCR, web crawling and sub-agents: three helpers, and when to leave them in the drawer".

## Web crawling, search APIs and scraping

**In short.** Search means asking a question and getting ranked pages; crawl means walking a whole site; scrape means pulling fields off one page (plan section 8). The real risk is a site's terms of service, the contract you accept by using it, not the law (plan section 8). Every robot will honor robots.txt, a site's public list of pages robots may visit, and every row will carry a license class, a label saying what you may do with it (plan section 8).

#### Comment

**What it means for you:** Any row in the database will show where it came from and what you may do with it.

**Decision needed:** none, already decided in the plan.

**Money:** none.

**When:** M2, foundation build, me, days 3 to 7 after go, when the database is built (plan section 5).

**Read more.** 4-the-guide.pdf, "Chapter 9. Data sources: where every number comes from, and what you may do with it".

## Sub-agents

**In short.** A sub-agent is a helper AI session with its own working memory, called context, that returns a summary (plan section 8). Helpers suit parallel reading, not numbers you must audit, so helpers write rows, not prose (plan section 8). Cursor's cloud agents provide this when needed; the plan defers it to v2 (plan section 8).

### Comment

**What it means for you:** No sub-agents in the first build; every number will come from one traceable step.

**Decision needed:** none, already decided in the plan.

**Money:** none.

**When:** not before M4. Plan section 8 says v2, months 3 to 6; plan sections 4 and 5 say M5, only if others ever use it (plan sections 2c, 4, 5, 8).

**Read more.** 4-the-guide.pdf, "Chapter 10. OCR, web crawling and sub-agents: three helpers, and when to leave them in the drawer".

## Benchmark literacy

**In short.** A benchmark is a public exam for AI models. The Intelligence Index is a weighted score across 10 tests; the Coding Agent Index scores a model plus its harness, the tool around it; AA-Omniscience measures whether a model makes things up (plan section 8). Tokens per second is typing speed; time to first token is the wait before it starts; cost per task is the dollars per question, the number that maps to your bill (plan section 8).

### Comment

**What it means for you:** Two traps: reasoning tokens bill as output, and one model has many hosts at different prices; pick per job, not by headline (plan section 8).

**Decision needed:** none, already decided in the plan.

**Money:** OpenRouter \$30 to 70 per month, spend limit \$75 (plan section 6).

**When:** M1, accounts and Cursor, you, 3 evenings, when you set up OpenRouter and Cline, a free coding extension you use as a test bench (plan section 5).

**Read more.** 4-the-guide.pdf, "Chapter 7. OpenRouter and benchmark literacy: one door to every model, and how to read the scoreboard".

## Figma and FigJam

**In short.** Figma is a design tool and FigJam is its whiteboard. Figma exposes a connector, a link this app uses while signed in as you, so I create FigJam boards, the system diagrams, in your account (plan section 8). You open them like any Figma file and drag boxes; you only draw a screen when you build one, which the plan calls v3 (plan section 8).

### Comment

**What it means for you:** The diagrams are yours to edit, and moving a box changes no code or number.

**Decision needed:** none, already decided in the plan.

**Money:** none.

**When:** M0, documents and diagrams, me, done today 2026-09-14; your own screens wait for a front end, which plan section 5 places in M5, only if others ever use it, later.

**Read more.** 4-the-guide.pdf, “Chapter 13. Figma and FigJam: the whiteboard where the system is drawn”.

## Fine-tuning at three levels

**In short.** Fine-tuning usually means retraining a model on your own examples; the plan uses the word at three levels (plan section 8). Level 1 is the harness, meaning the rules file, the MCP tools (a standard way to hand a model tools), the prompts and the model routing; it costs \$0, gives 95% of the quality, and is what you will do (plan section 8). Level 2 is better prompts and context. Level 3 trains a model as an outsourced service, about \$2 per run for 800 examples, but it changes form, not facts, and needs 500 to 1,000 hand-corrected examples (plan section 8).

### Comment

**What it means for you:** Your fine-tuning is editing rules and prompts in Cursor; level 3 is not in v1 and returns only with 500+ logged corrections and a repeated formatting failure (plan section 8).

**Decision needed:** none, already decided in the plan.

**Money:** level 1 \$0 (plan section 8); a level 3 GPU endpoint, a rented graphics computer, left running can eat \$200 in a weekend (plan section 8), which is the whole monthly ceiling (plan section 6).

**When:** M2, foundation build, me, days 3 to 7 after go, for the first harness; M3, you take the wheel in Cursor, weeks 2 to 4, when you edit it (plan section 5).

**Read more.** 4-the-guide.pdf, “Chapter 8.”Fine-tuning” at three levels: the harness, the context, and the model”.

## Section 9: Non-goals, risks, cut lines, deferred

### READ FIRST

**Read the risks and the cut lines.** Understand which checks protect the work and which features can wait if the schedule changes.

## Non-goals for v1

**In short.** A non-goal is something the plan leaves out on purpose. “v1” is the first working version, which M2 through M4 will build. Plan section 9 lists these non-goals:

- Valuing your own holdings or tracking your P&L, meaning your personal profit and loss.
- Real-time prices, meaning prices that refresh every second.
- Multi-user use, meaning more than one person logging in, and any commercial use.
- Any buy or sell output of any kind.
- Scraping Bloomberg, INDmoney, screener.in or Tickertape. Scraping means a program copying data off a website.
- Docker or self-hosting, meaning running the system on your own packaged server.
- Python in the production path, the code that actually runs the nightly jobs. That code stays in TypeScript.
- The option-value sheet and the lease and R&D converter sheets, extra workbook tabs for special cases.
- Fine-tuning any model, meaning retraining a model on your own data.

- BYOK inside Cursor. BYOK means “bring your own key”, pasting your own model key into Cursor.

#### Comment

**What it means for you:** if you catch yourself wanting one of these, the answer is “not in v1”. Leaving them out keeps the build small enough to finish.

**Decision needed:** none, already decided in the plan.

**Money:** none. Cursor Pro stays at \$20 per month, already paid, with no extra keys inside it (plan section 6).

**When:** M0 to M4. Some come back only in M5, only if others ever use it, later.

**Read more.** 1-read-this-first.pdf, 1. *What we are building, in one page.* 4-the-guide.pdf, Chapter 9. *Data sources: where every number comes from, and what you may do with it.*

## Risks and what catches them

**In short.** A risk is a thing that could go wrong. A mitigation is the catch that stops it from hurting. Plan section 9 pairs each risk with its catch, one sentence each below.

- If the code does the same math in a different order and breaks the 1e-6 golden test, checks inside each module, a 1e-9 warning and a later second golden case catch it (plan section 9). The golden test checks that the engine reproduces the Almarai example to within one millionth.
- If the free Neon database, the service that holds the numbers, fills its 0.5 GB limit, moving to the paid Launch tier at M4 catches it (plan section 9).
- If Pinecone, the search index for Damodaran’s writing, bills for re-sending unchanged text, a content-hash manifest catches it (plan section 9). That manifest is a list of fingerprints, one per text, so unchanged text is never re-sent.
- If a GitHub Actions job, the robot that runs jobs, hits the 6-hour cap or a top-of-hour schedule is dropped, chunked jobs, off-hour timers and a staleness alarm catch it (plan section 9). A staleness alarm fires when the last run is too old.
- If a licensed number leaks to a place it may not go, four locks catch it: NOT NULL, a database CHECK, a retrieval allow-list and `data_collection: deny` (plan section 9). NOT NULL means the database refuses a blank source or license, CHECK means it refuses a wrong value, and an allow-list means the search reads only approved sources, and `data_collection: deny` is a setting that tells crawlers and tools not to collect that text.
- If the corpus, the collection of his writing, has gaps in blog years 2012 and 2025 or in image-only decks, backfill and OCR before tier 1 is declared complete catch it (plan section 9). OCR turns images of text into text, and tier 1 is the blog posts.
- If you become dependent on the AI tool, attempt-then-ask, owning the tests, one no-AI evening a week and a decisions log catch it (plan section 9).
- If Cursor’s included model pool runs out, coding falls back to Composer or Grok, and the pipeline is unaffected because it never used Cursor’s models (plan section 9).

#### Comment

**What it means for you:** every risk has its catch written down before the build starts. You do not have to remember them, and I will build most of the catches.

**Decision needed:** none, already decided in the plan.

**Money:** Neon \$0, later about \$10 to 20 for the Launch tier; Mistral OCR about \$10 one-off for the image-only decks (plan section 6).

**When:** most catches land in M2, foundation build, me, days 3 to 7 after go. The Neon move waits for M4, weeks 4 to 8. The no-AI evening starts in M3, weeks 2 to 4.

**Read more.** 4-the-guide.pdf, *Chapter 0. How to learn with an AI coding tool without becoming dependent on it* and *Chapter 3. GitHub Actions: the robot that tests your code and runs it at night.*

## Cut lines

**In short.** A cut line is a rule for what to drop if time runs short. Plan section 9 sets one rule and one “never cut” list. The rule: if month 3 arrives and M4 is unfinished, ship US-only and postpone EODHD, the paid feed for Indian company numbers (plan section 9). India without its fallback tests would be worse than no India. The “never cut” list has six items (plan section 9): the golden test in CI, three vintage ids on every run, source and license NOT NULL, the India fallback log, the no-advice test, and cited claims in every memo. CI means continuous integration, the automatic test run on every change. A vintage id names which dated version of a Damodaran table a run used. The India fallback log records when an industry number came from emerging markets or Global instead of India.

### Comment

**What it means for you:** if the schedule slips, the India feature waits, not the safety checks. You will still have a working US system.

**Decision needed:** none, already decided in the plan.

**Money:** postponing EODHD saves \$60 per month, a line that only starts from M4 anyway (plan section 6).

**When:** checked when month 3 arrives after go (plan section 9), which is after the planned end of M4, weeks 4 to 8.

**Read more.** 1-read-this-first.pdf, 3. *The six rules that make the numbers trustworthy.* 4-the-guide.pdf, *Chapter 16. The schedule: twelve evenings-only weeks, from “go” to a working system.*

## Deferred

**In short.** Deferred means “not now, maybe later”. Plan section 9 lists these:

- pi, an optional terminal tool, where a terminal is the text window for typing commands. pi may come later, or never.
- Sub-agents, helper AI programs that run tasks for the main one.
- Blacksmith, a faster paid runner for CI jobs.
- The four remaining regional tables and indname.xls, the file that maps company names to industries (plan section 9).
- Option value and the converter sheets.
- A second golden case, Apple TTM. TTM means trailing twelve months, the last four quarters added up.
- Parallel or Exa news discovery, two services that find news stories.
- A Pinecone Assistant spike, a short trial of Pinecone’s built-in question tool.
- A web app, and commercial licenses, the paid data rights needed if others ever use the system.

### Comment

**What it means for you:** none of these block the first working system. You will not need to learn them in M1 to M4.

**Decision needed:** none, already decided in the plan.

**Money:** none now. Parallel about \$18 is optional and only if it is ever used (plan section 6).

**When:** after M4, weeks 4 to 8, at the earliest. Several belong to M5, only if others ever use it, later.

**Read more.** 4-the-guide.pdf, *Chapter 10. OCR, web crawling and sub-agents: three helpers, and when to leave them in the drawer.* 3-the-other-side.pdf, *The Other Side, Chapter 6: What This Means for the App.*

## Section 10: Technical appendix, in plain words

### OPTIONAL - FIRST PASS

**Technical reference for the build.** Keep this for inspecting formulas, tables, jobs, and interfaces after you understand the rest of the plan.

### 10.1 Correctness rules

**In short.** Five rules keep every number honest. Each will be enforced by a test or a database rule, not by memory.

- **Golden test.** The engine must reproduce the workbook's own example, Almarai, to within 1e-6, one millionth (plan section 10.1). That example gives 7.187840270062114 per share against a price of 72.28 (plan section 10.1, the verified golden case). A script will pull the expected numbers from the workbook, and CI, the robot that runs tests on every change, will lock that file so no AI agent can edit the test instead of the code (plan section 10.1).
- **Vintage.** A vintage is the dated version of one data file. Every run will store three vintage ids: market, country risk, industry (plan section 10.1). The 2026 data alone holds risk-free rates of 3.95%, 4.58% and 4.75%, and ERPs of 4.20%, 4.23% and 4.46% (plan section 10.1). ERP is the equity risk premium, the extra return people demand for holding stocks instead of safe bonds.
- **Source and license class.** Every stored fact will name its source and one of seven license classes, and the librarian's index will refuse the last three: personal licensed data, proprietary notes, link-only sources (plan section 10.1).
- **India rules.** The rupee risk-free rate is the 10-year government bond yield published by the RBI, India's central bank, minus India's default spread, so country risk is not counted twice (plan section 10.1). India's ERP is the mature ERP plus the country risk premium: 4.23% plus 2.85% equals 7.08% for January 2026 (plan section 10.1). An industry with fewer than 10 Indian firms falls back to emerging markets, then Global; 27 of 94 Indian industries are that thin (plan section 10.1).
- **No advice.** A test will scan every report, memo and chat answer for words that recommend a trade or name a future price (plan section 10.1).

#### Comment

**What it means for you:** You cannot break these rules by accident; the code and the database will refuse the wrong input.

**Decision needed:** none, already decided in the plan.

**Money:** none.

**When:** M2, foundation build, me, days 3 to 7 after go, for the first three rules and the advice test; M4, weeks 4 to 8, for the India rules (plan section 5).

**Read more.** 1-read-this-first.pdf, section “3. The six rules that make the numbers trustworthy”. 4-the-guide.pdf, “Chapter 9. Data sources: where every number comes from, and what you may do with it”.

## 10.2 The engine

**In short.** The engine is the calculator that turns inputs into a value per share. It will be written in TypeScript as pure functions. A pure function always returns the same output for the same input and changes nothing else, like a calculator with no memory. It will use float64, the standard 64-bit format computers use for decimal numbers. A module is one code file with one job, and each module mirrors one workbook sheet (plan section 10.2):

| File                    | What it computes (plan section 10.2)                                                                                                                                                                                                           |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rating.ts               | a credit rating estimated from interest coverage, with three firm-type tables and the country default spread                                                                                                                                   |
| costOfCapital.ts        | the cost of capital by four approaches; two of them add the current risk-free rate minus the rate embedded in the table                                                                                                                        |
| fcff.ts and terminal.ts | the cash flow path: growth, margin convergence, tax fade, reinvestment with the lag switch, past losses, discount factors, terminal cost of capital as risk-free plus mature ERP, terminal reinvestment as growth divided by return on capital |
| failure.ts              | the failure adjustment                                                                                                                                                                                                                         |
| options.ts              | the option adjustment, switched off for Almarai                                                                                                                                                                                                |
| india.ts                | the India rules from 10.1                                                                                                                                                                                                                      |
| reverseDcf.ts           | the growth or margin the current price implies (plan section 5)                                                                                                                                                                                |
| monteCarlo.ts           | ranges from random draws over the industry quartiles (plan section 5)                                                                                                                                                                          |
| impliedErp.ts           | his goal-seek as a root finder, a routine that adjusts one input until an output hits a target; anchors: index 7398.084728 at ERP 4.25%, expected return 8.8437%                                                                               |

Every output carries an audit: rating basis, missing interest flag, tier used, three vintage ids, sources, engine version and a hash, a short fingerprint, of the inputs (plan section 10.2). Excel and JavaScript both use float64, so an order-faithful port agrees to about 1e-12 (plan section 10.2).

### Comment

**What it means for you:** You will be able to open one file, hold the matching sheet beside it, and check the formulas line by line.

**Decision needed:** none, already decided in the plan.

**Money:** none.

**When:** M2, foundation build, me, days 3 to 7 after go (plan section 5).

**Read more.** 2-damodaran-essentials.pdf, “Chapter 3: The DCF Chain, Exactly As the Workbook Does It”. 4-the-guide.pdf, “Chapter 4. TypeScript for a Python brain”.

## 10.3 The database tables

**In short.** The database will live on Neon, a hosted version of Postgres. A table is a grid with named columns, one row per record. NOT NULL means a column may never be empty. CHECK means a value must pass a test before it is stored. The plan names twenty-one tables, grouped here by purpose (plan section 10.3).

| Purpose                       | Tables                                                                       | Rules worth knowing                                                                                                   |
|-------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Reference data from his files | vintage, country_risk, industry_stat, industry_dist, erp_monthly, erp_annual | firm counts and quartiles kept; his ERP columns copied verbatim                                                       |
| Companies and facts           | company, fact, ltm_financials, watchlist                                     | in fact, source, license class, filing date, accession and vintage id are all NOT NULL; missing interest is a flag    |
| Memos and claims              | memo, claim, valuation_inputs, narrative_note                                | memos are versioned, base or opposing; every input row points to a claim                                              |
| Runs and results              | valuation_run, realised_price, market_context_event                          | a run stores input hash, outputs, three vintage ids NOT NULL, tier, rating basis, seed; headlines are annotation only |
| Jobs and evals                | job_heartbeat, rag_eval_case, rag_eval_run, chunk_manifest                   | chunk_manifest carries the license CHECK that refuses the three forbidden classes                                     |

Raw files sit in a private R2 bucket, a file store on Cloudflare, keyed by sha256, a fingerprint of the file's bytes (plan section 10.3).

### Comment

**What it means for you:** The database will refuse a number without a vintage, so the honesty rules live in the storage.

**Decision needed:** none, already decided in the plan.

**Money:** Neon \$0 now, later about \$10 to 20 per month; Cloudflare R2 \$0 (plan section 6).

**When:** M2, foundation build, me, days 3 to 7 after go (plan section 5).

**Read more.** 4-the-guide.pdf, “Chapter 5. Postgres and Neon: the one place the numbers live”.

## 10.4 The librarian and the models

**In short.** The librarian is the search part of the system. It will find Damodaran passages that answer a question and hand them to the model with citations.

- Vectors and embeddings. An embedding turns text into a vector, a list of numbers that captures its meaning. Voyage's voyage-context-4 makes vectors of 1024 numbers, stored in a Pinecone serverless index (plan section 10.4).
- Reranker. A second model that re-sorts the first results; rerank-3 cuts 50 candidates to 8 (plan section 10.4).

- **Metadata.** The label on each stored chunk: source, title, date, document type, license class, vintage, speech flag, region, company (plan section 10.4).
- **Tiers.** Batches indexed in order: tier 1 the blog; tier 2 the book, packets with speaker notes, exams; tier 3 dataset rows as sentences; tier 4 transcripts after a punctuation pass of about \$11 (plan section 10.4). Third-party press, filings, the Tykoh guide, Bloomberg and EODHD are never indexed (plan section 10.4).
- **Router.** The rule that picks a method: a question about one document pastes that document in; a question across the corpus searches (plan section 10.4).
- **Pinned model ids.** An exact model name fixed in code, recorded with the benchmark version it was chosen under. The four: `z-ai/glm-5.3` for memos and answers, `z-ai/glm-5.3-flash`: batch for bulk work, `moonshotai/kimi-k3` as judge, `deepseek/deepseek-v4.1-flash` as fallback (plan section 10.4). Prices are re-checked monthly; GLM-5.3 was listed at \$0.92 to 1.40 per million tokens in and \$3.14 to 4.40 out in September 2026 (plan section 10.4).

#### Comment

**What it means for you:** Every answer will cite its passage, and nothing paid or proprietary can leak into the index.

**Decision needed:** none, already decided in the plan.

**Money:** OpenRouter \$30 to 70 per month, first purchase \$20 of credits, spend limit \$75; Pinecone \$0 now, later \$20; Voyage \$0 (plan section 6).

**When:** M2, foundation build, me, days 3 to 7 after go, for tiers 1 and 3 (plan section 5); the plan sets no date for tiers 2 and 4.

**Read more.** 4-the-guide.pdf, “Chapter 6. Pinecone, Voyage and RAG in plain words” and “Chapter 7. OpenRouter and benchmark literacy: one door to every model, and how to read the scoreboard”.

## 10.5 The memo writer

**In short.** The memo writer will turn a filing and his writing into a story with numbers attached. It follows his five steps: develop the narrative; test it against history and common sense; convert it into value drivers; connect it to a valuation; keep the feedback loop open (plan section 10.5). Every claim passes the 3P gate, his test of whether a story is possible, plausible and probable.

- **Template.** The workbook’s “Stories to Numbers” sheet, one row per driver, each with a required link to the story (plan section 10.5).
- **Reverse DCF first.** A DCF, discounted cash flow, is the valuation itself; running it in reverse means starting from today’s price and asking what it assumes. Hold margin and cost of capital, solve for the year-10 revenue the price implies, turn it into market share, test that with 3P (plan section 10.5).
- **Guard rails.** An uncited claim rejects the memo. Numbers go into `valuation_inputs`, never read out of prose. The engine is never edited. You approve a diff, proposed against prior inputs, in Cursor. An opposing memo is mandatory. A re-run labels the change tweak, shift or break (plan section 10.5).
- **Plausibility bands.** A cost of capital outside 5.3 to 9.9% for the US, or 6.3 to 11.7% globally, is flagged (plan section 5).
- **Sources.** His 2014 narrative posts, the Uber exchange with Gurley, Corporate Life Cycle sessions 10 to 13, Data Update 5 for 2026 (plan section 10.5).

**Comment**

**What it means for you:** The model proposes, you approve, and the calculator never sees a number you have not seen.

**Decision needed:** none, already decided in the plan.

**Money:** OpenRouter \$30 to 70 per month, spend limit \$75 (plan section 6).

**When:** M2, foundation build, me, days 3 to 7 after go (plan section 5).

**Read more.** 2-damodaran-essentials.pdf, “Chapter 1: Why value, the story, and the 3P test”. 3-the-other-side.pdf, “The Other Side, Chapter 6: What This Means for the App”.

## 10.6 The jobs

**In short.** A job is a script that GitHub Actions, GitHub’s robot, runs on a schedule or on request. Cron is the schedule format. Dispatch means a run you start by hand. A heartbeat is a row a job writes when it finishes, so silence can be noticed. An accession is the id of one SEC filing. The nine jobs (plan section 10.6):

- load-damodaran: dispatch; loads one vintage of his datasets.
- ingest-edgar: nightly; pulls a filing only when a new accession appears.
- value-watchlist: nightly; revalues every watchlist name.
- watch-erp-crp: monthly and quarterly; checks his file names, and a new sha256 creates a vintage row and a GitHub issue.
- rag-index: dispatch; indexes one tier.
- rag-eval: nightly; scores the librarian, report only.
- news-headlines: nightly; pulls headlines from licensed sources.
- realised-prices: weekly; records later prices for the scoreboard.
- report: on demand; produces the markdown and PDF report.

Every job writes a heartbeat, and a staleness alarm fires after 36 hours (plan section 10.6). Schedules sit off the hour, because top-of-hour cron runs get dropped (plan section 5).

**Comment**

**What it means for you:** The robots work at night, and a missed night shows up as an alarm rather than a stale number.

**Decision needed:** none, already decided in the plan.

**Money:** GitHub CI \$0 to 25 per month (plan section 6).

**When:** M2, foundation build, me, days 3 to 7 after go, for the nightly robots; M4, weeks 4 to 8, for the first scored horizon (plan section 5).

**Read more.** 4-the-guide.pdf, “Chapter 3. GitHub Actions: the robot that tests your code and runs it at night”.

## 10.7 The market-regime panel

**In short.** A regime is the mood of the whole market. This panel will describe that mood and will never feed a number into a valuation.

- Implied ERP. His monthly headline figure and its variants from ERPbymonth: trailing 12 months, adjusted risk-free, smoothed, normalized, net cash yield (plan section 10.7).

- ERP against history. His histimpl series, 1960 to 2025, with his benchmark averages of 4.25% long run, 5.16% for 2006 to 2025 and 5.00% for 2016 to 2025 (plan section 10.7).
- Expected return on stocks. He calls 8 to 9% healthy (plan section 10.7).
- Earnings yield against bond yield. Context only, because he calls PE noisy and unreliable (plan section 10.7).
- Credit spreads. High-yield and BBB spreads from FRED, the US Federal Reserve's public data service, series BAMLH0A0HYM2, BAMLC0A4CBBB, DGS10; a spread is the extra yield on riskier bonds (plan section 10.7).
- Concentration. The weight of the top 7 stocks in the index, 30.9% in his 2026 update (plan section 10.7).
- Fair-value grid. What the index would be worth at each ERP from 2 to 6% (plan section 10.7).

Each indicator becomes a percentile band, the share of past readings below today's, with a plain sentence, for example that the market is pricing in less risk than its 10-year median (plan section 10.7). His caveat is printed on the panel in his own words: "I am not a market timer", he says, and he values the market at regular intervals to measure what it is pricing in, not to forecast where it goes (plan section 10.7). Headline sentiment stays in market\_context\_event as an overlay only and never enters a computation (plan section 10.7).

#### Comment

**What it means for you:** You will see what the market is pricing in, in plain sentences, with his own warning against timing beside it.

**Decision needed:** none, already decided in the plan.

**Money:** none.

**When:** M4, weeks 4 to 8; done when the panel reproduces his September 2026 numbers, ERP 4.09% and expected return 8.84% (plan section 5).

**Read more.** 2-damodaran-essentials.pdf, "Chapter 6: Relative Valuation and What the Market Prices In". 3-the-other-side.pdf, "The Other Side, Chapter 2: His Misses with the Numbers".

## Critical files

**In short.** Four locations hold everything the build depends on (plan section 10, Critical files):

- ~/Downloads/fcffsimploginzu.xlsx: the workbook and its golden case.
- ~/Downloads/damodaran-data-01.zip and -02.zip: the 2026 datasets, loaded once into Neon.
- financeMD/: the corpus of blog, packets, exams and book, including pc/implprem/ERPbymonth.md and pc/datasets/histimpl.md, for the librarian, the regime panel and the document the plan calls doc C.
- ~/code/valuation/packages/{engine,db,rag,memo,mcp-server,models}/, cursor/ and AGENTS.md: what M2 will create.

#### Comment

**What it means for you:** Keep the three Downloads items where they are until M2 has loaded them; the fourth path will not exist until I build it.

**Decision needed:** none, already decided in the plan.

**Money:** none.

**When:** M2, foundation build, me, days 3 to 7 after go (plan section 5).

**Read more.** 4-the-guide.pdf, “Chapter 15. The Damodaran watch and read order for this project”.  
1-read-this-first.pdf, section “2. Where the pieces live and who runs them”.